EE2005 ELECTRICAL DEVICES AND MACHINES

TUTORIAL 8 – Induction Motors 1

Q1. A 220-V, 3-phase, 6-pole, 50-Hz induction motor is running at a slip of 3.5 percent. Determine:

a. The speed of the rotating field in rev per min.
b. The speed of the rotor in rev per min.
c. The slip speed of the rotor.
d. The rotor frequency (i.e. the frequency of the voltage induced in the rotor) in hertz.

[Ans : (a) 1000 rpm (b) 965 rpm (c) 35 rpm (d) 1.75 Hz]

Q2. The air-gap power of a 6-pole, 60-Hz induction motor running at a speed of 1100 rpm is 5 kW. Determine the power developed?

[Ans : 4583.35 W]

Q3. A 3-phase, 8-pole, 50-Hz induction motor is running at a speed of 710 rpm with an input power of 35 kW. The stator copper loss at this operating condition is known to be 1200 W while the rotational losses are 600 W. Determine:

a. The rotor copper losses.
b. The power developed.
c. The output power.

[Ans : (a) 1802.554 W (b) 31997.446 W (c) 31397.446 W]

Q4. A 3-phase, 440-V, 50-Hz, 4-pole, Y-connected induction motor has a stator resistance of 0.12 ohm/phase. The rotational losses of the motor are 2.4 kW. At full-load, the motor takes an input power of 42.5 kW at a power factor of 0.85 lagging and runs at 1470 rpm. Determine:

a. The input current and stator copper losses.
b. The air-gap power and rotor copper losses.
c. The developed torque, load torque and efficiency of the motor.

[Ans : (a) 65.608 A, 1549.587 W (b) 40950.413 W, 819.008 W,
(c) 260.698 N-m, 245.108 N-m, 88.78 %]

Q5. A 4-pole, 440-V, 50-Hz, Y-connected, 3-phase induction motor has a stator impedance of (0.15 + *j* 0.8) ohm/phase and a rotor impedance (referred to the stator) of (0.2 + *j* 0.8) ohm/phase at standstill. The magnetizing reactance is 30 ohms/phase. The rotational losses of the motor are 750 W. Using the equivalent circuit, determine the following when the motor operates at a full-load slip of 4 %.

a. Input current and input power factor.
b. Air-gap power and developed power.
c. Load torque and efficiency of the motor.

[Ans : (a) 47.841 A, 0.899 lag (b) 31733.978 W, 30464.619 W
(c) 197.051 N-m, 90.693 %]

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TUTORIAL 9 – Induction Motors 2

Q1. A 3-phase, 440-V, 50-Hz, 4-pole, Y-connected induction motor has the following per-phase parameters referred to the stator.

$$R_1 = 0.4\ \Omega, R_2 = 0.5\ \Omega, X_1 = 1.2\ \Omega, X_2 = 1.5\ \Omega, X_M = 80\ \Omega$$

a) Determine the speed at which maximum torque is developed.
b) Determine the corresponding maximum torque developed, input current and input power factor.
c) Determine the starting torque developed.
d) What resistance should be added in each phase of the rotor circuit, referred to the stator, to obtain maximum starting torque developed?

[Ans : (a) 1223.4 rpm (b) 192.931 N-m, 62.217 A, 0.737 lag
(c) 74.825 N-m (d) 2.212 ohms]

Q2. A 3-phase, 220-V, 50-Hz, 4-pole, Y-connected induction motor is tested with the following results:

No-load : 220 V, 5.7 A, 380 W

Locked rotor : 79.5 V, 18.57 A, 675 W

DC test : 9.02 V, 17.2 A

a) Determine the per-phase parameters of the induction motor equivalent circuit referred to the stator and the rotational losses of the motor. Assume that $X_1 = X_2$.
b) Determine the efficiency of the induction motor when it is running at it's rated speed of 1410 rpm.

[Ans : (a) $R_1 = 0.262\ \Omega, R_2 = 0.39\ \Omega, X_1 = 1.192\ \Omega, X_2 = 1.192\ \Omega,$
$X_M = 21.091\ \Omega,\ P_{rot} = 354.463\text{W}$ (b) 83.538%]

Q3. Consider the induction motor with parameters as given in Q1. Obtain an expression for the developed power P_{dev} in terms of the slip *s* and hence determine the speed at which maximum power is developed by the motor and the value of the maximum developed power. Compare the answers with the answers obtained in Q1 for the power developed and the speed at which the maximum torque is developed.

$$[\text{Ans}: P_{dev_{\max}} = 25295.039 \text{ W}, n_m = 1274.55 \text{ rpm}]$$

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TUTORIAL 10 – Induction Motors 3

Q1. A 3-phase, 460-V, 60-Hz, 4-pole Y-connected induction motor has a full-load speed of 1740 rpm and the per-phase parameters referred to the stator are as follows.

$$R_1 = 0.25\ \Omega, R_2 = 0.2\ \Omega, X_1 = X_2 = 0.5\ \Omega, X_M = 30\ \Omega$$

The rotational losses of the motor are 1700 W. Determine

a) (i) The input current at starting.
   (ii) Starting torque developed.

b) (i) Full-load slip.
   (ii) Input current when running at full load.
   (iii) Ratio of the input starting current to the full-load input current.
   (iv) Input power factor at full load.
   (v) Full-load torque developed.
   (vi) Motor efficiency at full load.

c) (i) Slip at which maximum torque of the motor is developed.
   (ii) Maximum torque developed.
   (iii) Ratio of maximum torque developed to full-load torque developed.

d) What resistance should be added in each phase of the rotor circuit, referred to the stator, to obtain maximum starting torque developed?

[Ans : (a) 244.10 A, 183.489 N-m (b) 0.0333, 42.787 A, 5.705, 0.941 lagging, 162.989 N-m, 87.244 % (c) 0.1955, 429.381 N-m, 2.634 (d) 0.823 ohms]

Q2. A certain 3-phase induction motor has a torque developed of 345 N-m when running at a slip of 0.01. The rotor copper loss at this slip is 5.69 kW. Assume that the rotational losses are negligible.

a) If the rotor circuit resistance is increased by 5 times, determine the torque developed if the slip is also assumed to be increased by 5 times to 0.05.
b) What is the rotor copper loss with the conditions in (a).

[Ans : (a) 345 N-m (b) 28.4 kW]

Q3. Suppose that the stator winding resistance $R_1 = 0\ \Omega$.

Using the induction motor equivalent circuit and its Thevenin equivalent circuit, show that

$$\frac{T_{dev_{\max}}}{T_{dev}} = \frac{s_{\max}^2 + s^2}{2 s_{\max} s}$$

where

$T_{dev_{\max}}$ is the maximum torque developed with corresponding slip $s_{\max}$, and

T_{dev} is the torque developed at any slip s.

[The above equation shows the relationship between the torque developed at any speed and the maximum torque developed in terms of their respective slip values, if $R_1 = 0\ \Omega$.]

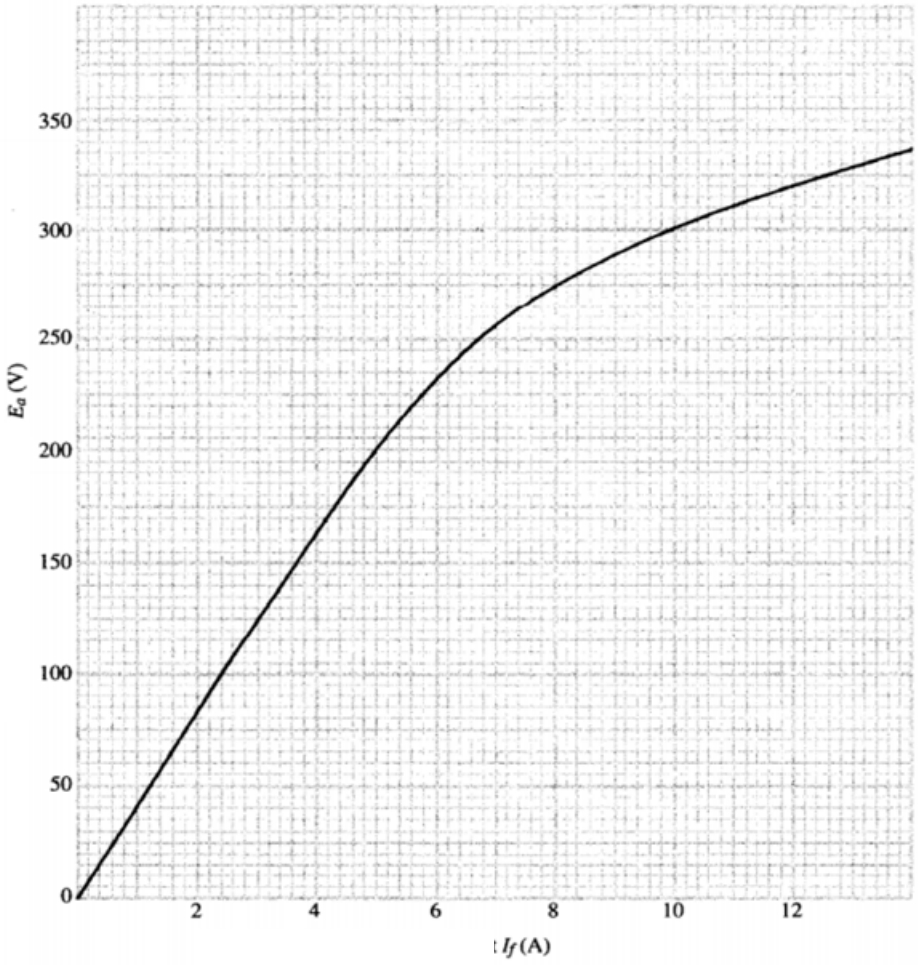
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TUTORIAL 11 - DC Motors

Q1. The magnetisation curve of a DC machine obtained at 1200 rpm is shown below. The armature winding resistance (including the brushes, interpoles and compensating windings) is 0.125 ohm and the field winding resistance is 15 ohms. The DC machine operates as a separately-excited motor with an armature terminal voltage supply of 250 V. Its field circuit is connected to a separate 120 V dc supply. The motor has a no-load speed of 1085 rpm. Determine

a) The rotational losses of the motor.
b) The speed, load torque and efficiency of the motor when it draws its rated current of 200 A from the supply.
c) The required armature terminal voltage when the motor runs at 800 rpm and its developed torque reduces to 70% of the value in part (b).

[Ans :(a) 2693.333 W (b) 981.818 rpm, 411.48 N-m, 83.019 % (c) 200.833 V]



Q2. A 240-V DC shunt motor has an armature winding resistance of 0.3 ohm and field winding resistance of 120 ohms. The rotational losses and armature reaction in the motor are negligible. At full load, the motor takes an input power of 10 kW and runs at 1000 rpm.

a) Determine the internal generated voltage and developed torque of the motor at full load.
b) Determine the armature current and torque developed by the motor when it is running at a speed of 990 rpm.
c) Determine the speed of the motor when a 60 ohms resistance is added in series with the field winding. Assume that the motor is running with 50% full load torque and is operating in the linear part of the magnetisation curve.

[Ans :(a) 228.1 V, 86.402 N-m (b) 47.27 A, 102.963 N-m
(c) 1519.931 rpm]

Q3. A 240-V DC shunt motor has an armature winding resistance of 0.25 ohm and rotational losses of 500 W. The field winding resistance is 120 ohms.

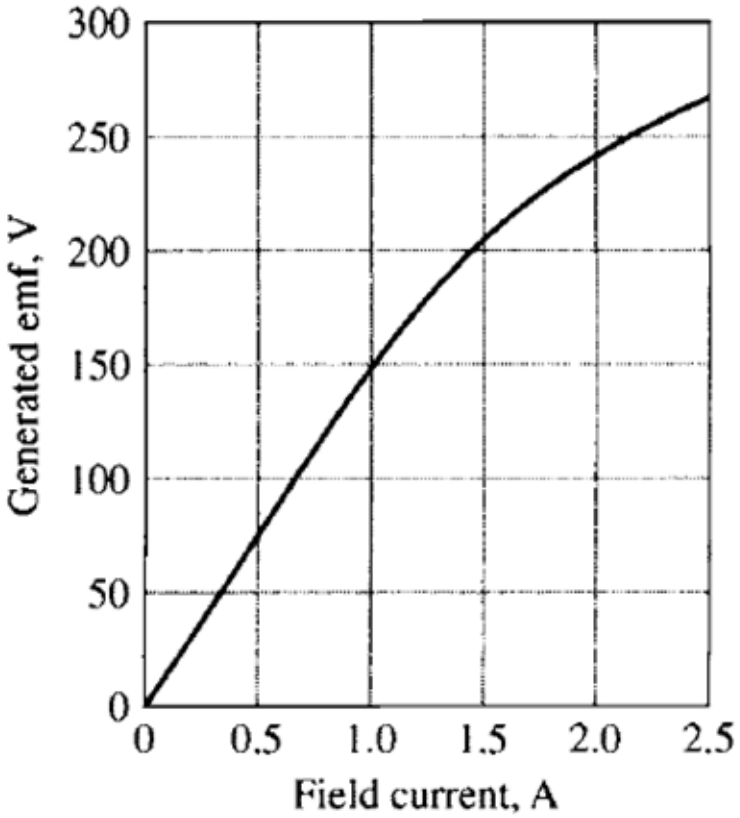
a) Determine the internal generated voltage, developed power, output torque and efficiency of the motor if it takes an input power of 20 kW at 240 V while running at 1200 rpm.
b) If the internal generated voltage is 230 V, determine the input current, motor speed, output power and developed torque of the motor.
c) What would be the input current and speed of the motor if the load is disconnected from the motor shaft?

[Ans :(a) 219.667 V, 17866.176 W, 138.196 N-m, 86.831 %
(b) 42 A, 1256.447 rpm, 8.7 kW, 69.922 N-m
(c) 4.088 A, 1308.224 rpm]

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TUTORIAL 12 - DC Generators

Q1. The magnetisation curve of a 250-V DC machine at a speed of 1200 rpm is given below. The machine is connected as a separately excited dc generator. The armature winding resistance of the machine is 0.14 ohm.



a) Suppose that the generator is supplying an armature current of 100 A. With the generator speed held at 1200 rpm, calculate the output power of the generator for constant field currents of

   (i) 1.0 A
   (ii) 2.0 A
   (iii) 2.5 A

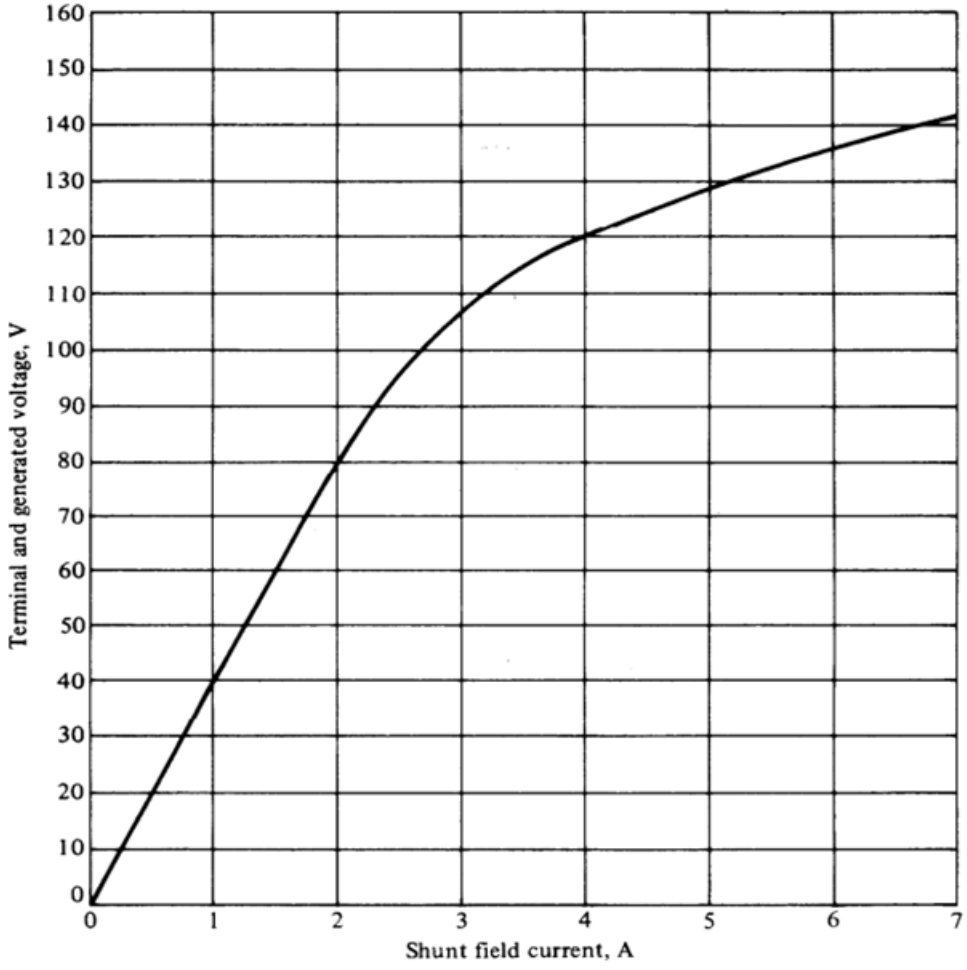
b) Repeat part (a) if the generator's speed is reduced to 900 rpm.

[Ans :(a) 13.6 kW, 22.6 kW, 25.6 kW (b) 9.85 kW, 16.6 kW, 18.85 kW]

Q2. The magnetisation curve of a 120-V, 10-kW, 1800-rpm DC machine is given below. The armature and field winding resistances of the machine are 0.20 ohm and 18 ohms, respectively. The field circuit has a rheostat of 0 to 15 ohms connected in series with the field winding. The rotational losses of the machine are 650 W. The DC machine operates as a separately-excited generator with a field circuit voltage of 130 V and it is driven at 1800 rpm. Calculate

a) The field rheostat setting to obtain a terminal voltage of 120 V at no-load condition.
b) The field rheostat setting when the generator supplies an output power of 9600 W at a rated terminal voltage of 120 V. Neglect armature reaction.
c) The efficiency of the generator for the above load condition.

[Ans :(a) 14.5 ohms (b) 3.667 ohms (c) 77.985 %]



Q3. The DC machine in Q2 is now operated as a shunt-connected DC generator and is driven at 1800 rpm. Determine

a) The field rheostat setting to obtain a terminal voltage of 120 V at no-load condition.
b) The terminal voltage when the generator delivers an armature current of 50 A. Neglect armature reaction. Use the field rheostat setting as obtained in (a).
c) The input shaft torque and efficiency of the generator for the load condition in (b).

[Ans :(a) 12 ohms (b) 105 V (c) 33.953 N-m, 76.289 %]